

Prompt Engineering

Mehmet Ali Akyol, PhD

Ankara Medipol University

December 5, 2025

Week 8: Evaluating AI Outputs

The Evaluation Gap

The Paradox of Generative AI

It is significantly easier to **generate** text than to **verify** it.

- **Generation:** Low effort, high speed, infinite variability.
- **Verification:** High effort, requires domain expertise, slow.
- **The Risk:** Without rigorous evaluation, we scale errors, not solutions.
- **The Goal:** Close the gap by automating verification where possible and focusing human effort where necessary.

Why Evaluation Matters

- **Reliability:** Moving from “it looks good” to “it works 99% of the time.”
- **Safety:** Detecting edge cases, jailbreaks, and harmful outputs before deployment.
- **Iterative Improvement:** You cannot improve what you cannot measure. Evaluation provides the signal for the refinement loop.
- **Model Selection:** Data-driven decisions on which model (GPT-4 vs. Llama 3) offers the best cost/performance ratio.

Weekly Learning Objectives

- Define evaluation criteria aligned with task goals.
- Apply qualitative and quantitative assessment methods.
- Design human-in-the-loop review pipelines.
- Use automation (judge models, checklists) responsibly.
- Interpret evaluation results to guide refinement.

Key Concepts

- Quality dimensions (accuracy, completeness, tone, compliance)
- Intrinsic vs. extrinsic evaluation
- Rubrics, scorecards, and annotation protocols
- Sampling strategies (random, risk-based, stratified)
- Inter-rater reliability and bias mitigation

Framework: The 3 H's of Evaluation

1. Helpful

- Does it solve the user's problem?
- Is the format correct?
- Is the tone appropriate?

2. Honest

- Is it factually accurate?
- Does it admit uncertainty?
- No hallucinations?

3. Harmless

- No toxicity or bias.
- No dangerous content.
- Resists jailbreaks.

- **Clarify Intent:** What is the “Job to be Done”? (e.g., Summarize vs. Analyze).
- **Define Failure Modes:** What does a *bad* response look like? (e.g., Missing citations, wrong JSON format).
- **Balance Metrics:**
 - *Hard Metrics:* Syntax, length, forbidden words (Deterministic).
 - *Soft Metrics:* Creativity, empathy, flow (Probabilistic).

Rubric Construction: Best Practices

- **Binary Checklists:** Best for strict constraints.
 - “Does the output contain a disclaimer?” (Yes/No)
- **Likert Scales (1–5):** Best for nuance.
 - 1 = Completely wrong.
 - 3 = Mostly correct but requires editing.
 - 5 = Perfect, ready to publish.
- **Comparative Ranking:** Best for preference.
 - “Is Output A better than Output B?”

The Golden Dataset (Ground Truth)

Definition

A curated set of inputs and their corresponding “perfect” outputs, used as a benchmark for testing.

- **Creation:** Hand-written by experts or verified high-quality model outputs.
- **Size:** Start small (10–50 examples) but high diversity.
- **Usage:** Run every prompt version against this set to detect regression.
- **Maintenance:** Update the dataset as new edge cases are discovered in production.

Qualitative Techniques (Human-Centric)

- **Expert Review:** Subject Matter Experts (SMEs) grade outputs. High cost, high accuracy.
- **Crowdsourcing:** Non-experts rate for general feel/fluency. Lower cost, variable quality.
- **Red Teaming:** Adversarial testing to find safety flaws.
- **User Feedback:** Thumbs up/down, “Regenerate” rate in production.

Quantitative Techniques (Metric-Centric)

- **N-gram Metrics (BLEU, ROUGE):** Measures word overlap. Good for translation, bad for reasoning.
- **Embedding Similarity:** Measures semantic closeness to the Golden Dataset using vector distance (Cosine Similarity).
- **Code Execution:** Does the generated code run? Does it pass unit tests?
- **Factuality Scores:** Using search engines to verify claims (e.g., RAGAS metrics).

Sampling Strategies

- Random sampling for baseline quality tracking.
- Risk-based sampling (e.g., high-stakes domains, new templates).
- Stratified sampling across personas, use cases, data segments.
- Event-driven sampling after major prompt or model changes.

- **Direct Scoring:** “Rate this response from 1 to 5 based on clarity.”
- **Pairwise Comparison:** “Which response is better, A or B?” (Often more reliable than scoring).
- **Reference-Based:** “Compare the response to this Golden Answer. Is it factually consistent?”
- **Rubric-Based:** “Does the response meet criteria X, Y, and Z? Answer Yes/No for each.”

Judge Models and Checklists

- **The Judge:** A stronger model (e.g., GPT-4) evaluates a weaker model (e.g., Llama-7B).
- **Deterministic Checks:** Regex for phone numbers, JSON validation, forbidden word lists.
- **Hybrid Pipelines:**
 - 1 Fast automated checks (Syntax).
 - 2 LLM Judge (Semantics).
 - 3 Human Review (Edge cases/Low confidence scores).

Dashboards and Reporting

- Track key metrics over time (accuracy, adherence, response time).
- Segment by task, persona, channel for targeted insights.
- Highlight trends, anomalies, and top failure modes.
- Share reports with stakeholders to drive accountability.

Common Evaluator Biases

- **Verbosity Bias:** Models (and humans) tend to rate longer answers as better, even if they are fluff.
- **Egocentric Bias:** LLMs prefer outputs generated by themselves or models from the same family.
- **Position Bias:** In pairwise comparison, the first option (Option A) is often preferred.
- **Fix:** Swap positions (A/B testing), force brevity penalties, and use blind grading.

Reducing Reviewer Bias

- **Blind Reviews:** Hide the model name/version from the human reviewer.
- **Calibration Sessions:** Have all reviewers grade the same 5 examples and discuss discrepancies.
- **Rotation:** Rotate reviewers to prevent fatigue and “anchoring” to a specific style.
- **Gold Standard Injection:** Slip known “bad” examples into the queue to check if reviewers are paying attention.

Inter-Rater Reliability

- Measure agreement (Cohen's kappa, Krippendorff's alpha).
- Investigate disagreements to refine rubrics.
- Provide refresher training and updated exemplars.
- Automate alerts when reliability drops below threshold.

- Criteria: accuracy, completeness, tone, compliance.
- Process: human review + policy checklist + judge model.
- Report: monthly dashboard highlighting clause omissions.
- Outcome: 30% reduction in manual escalations.

Marketing Campaign QA

- Criteria: brand voice, CTA clarity, localization quality.
- Use A/B testing metrics (CTR, conversion) for extrinsic evaluation.
- Sample high-risk markets at higher frequency.
- Share scorecards with creative team for continuous improvement.

- Criteria: factual accuracy, citation fidelity, coverage of sections.
- Workflow: human subject matter expert review + fact-check prompts.
- Use question-answer pairs to probe understanding.
- Publish evaluation logs with each delivered summary.

Key Takeaways

- Evaluation turns AI outputs into dependable deliverables.
- Combine qualitative judgment with quantitative tracking.
- Automate selectively; keep humans where nuance matters.
- Use evaluation insights to steer prompt and process refinements.

Thank You!

Dr. Mehmet Ali Akyol

mehmetali.akyol@gmail.com

Questions?